Activity A (i)

The minimum specification a computer would need to download/ view the Health Advice Group Website:

Operating system - Windows 8 or later, macOS High Sierra 10.13 or later, 64-bit Ubuntu 14.04+, Debian 8+, openSUSE 13.3+ or Fedora Linux 24+

Processor – Intel Pentium 4 or later

Memory - 2 GB minimum, 4GB recommended

Screen resolution - 1280x1024 or larger

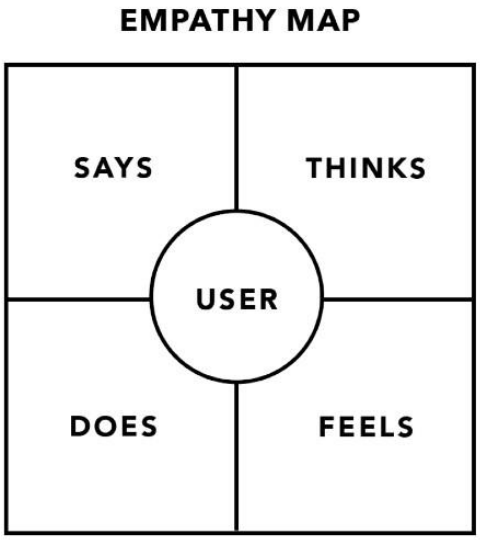
Application window size - 1020x680 or larger

Internet connection – Required Research:

APIs

A lot of the features requested from Health Advice Group include some form of weather related information, a way to find this information consistently and with high accuracy would be through the use of APIs, APIs in this case would act as an intermediary between the application and the information, allowing relevant data to be transmitted to the website in real time.

Empathy mapping



Later on after developing the a prototype for the application receiving feedback is an integral process for ensuring the quality of the service, in order to receive the best feedback possible the actions of the subject we are interviewing should be classified into 4 separate categories: says, thinks, does and feels, this approach looks beyond what someone simply says and therefore we receive a more genuine analysis into the quality of the final product.

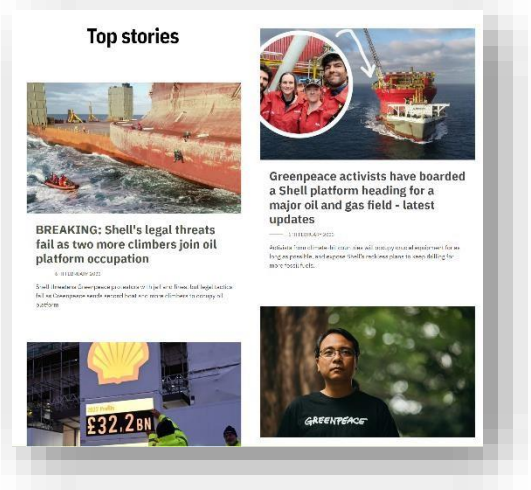
During the interviews there are four key questions that should be answered:

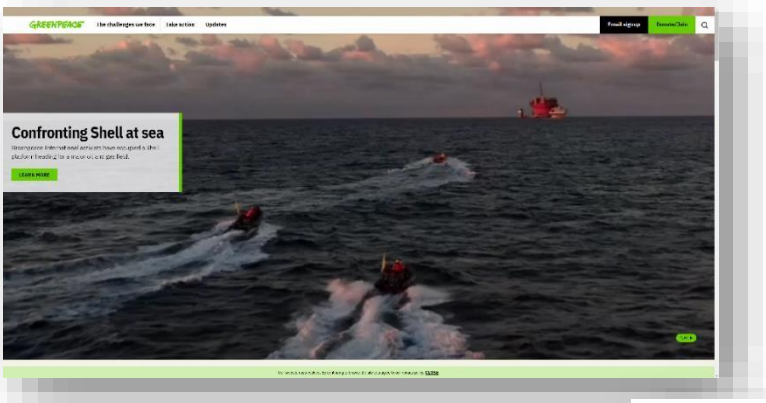
1. What is your overall impression of our service?
2. Would you recommend this product to a friend?
3. What changes would you like to see added to taken away?
4. How likely are you to use this application regularly?

<u>Appendix:</u>

GreenPeace

In order to get a better understanding of how to carry out developing the application for the Health Advice Group to the quality they require, it will be useful to compare against other pre-established charities in a similar field and notice any patterns between them. The first charity I looked at is called GreenPeace, who primary objective is to help protect and maintain the environment, the application they've developed employs a handful of features that can be easily found upon entering the page, the most prominent being the story of upmost relevance being displayed in the centre of the screen.



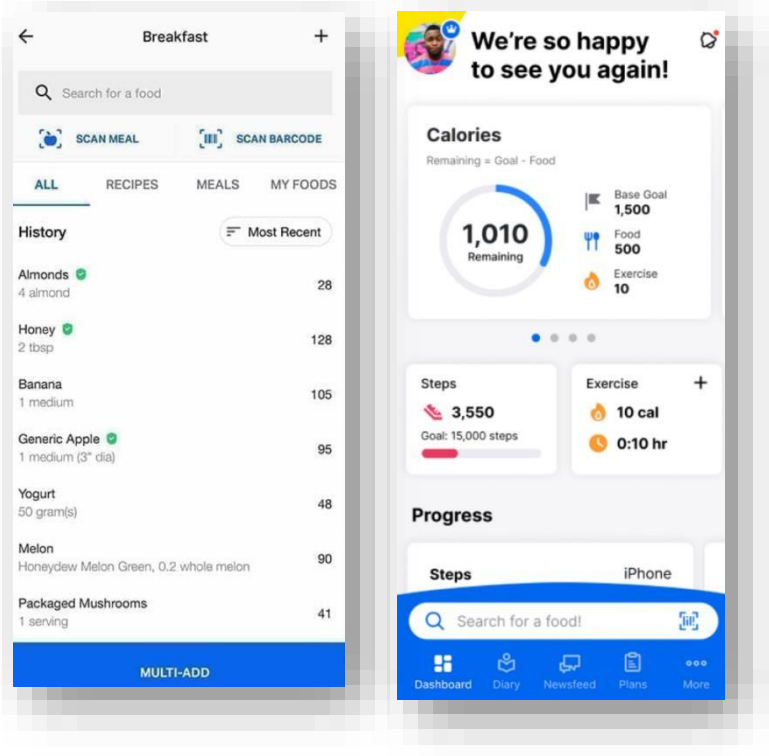


They also have a prominent green button for accepting donations allowing them to continue their cause as well as, by scrolling down, multiple stories about the ongoing fight against climate change in order to keep the reader informed and an important role in change.

Another feature Greenpeace has is the ability to sign up to their newsletter by clicking an easily distinguishable button so that users can make the choice to receive information even when they are not on the website itself.

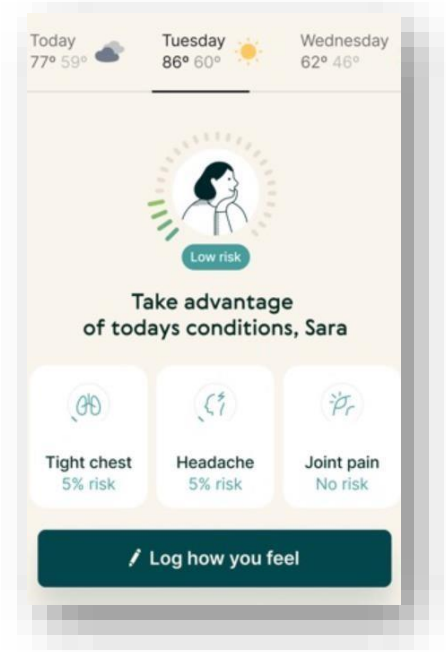
MyFitnessPal

This is a great example of an application focused solely towards the health of the individual user, it features a great way of tracking all the food you eat in terms of calories and other macros through an extensive database that the user can scroll through or search for a specific product and add it to their daily intake with ease. It also compares the amount of food they have already digested to the amount they have left to eat, this is great in helping make sure people don't skip meals and can stay on track towards their fitness goals. I plan to use the idea of having a database in my application that users can sort through in order to find specific items or foods as, I believe, it adds a sense of quality towards the final product.



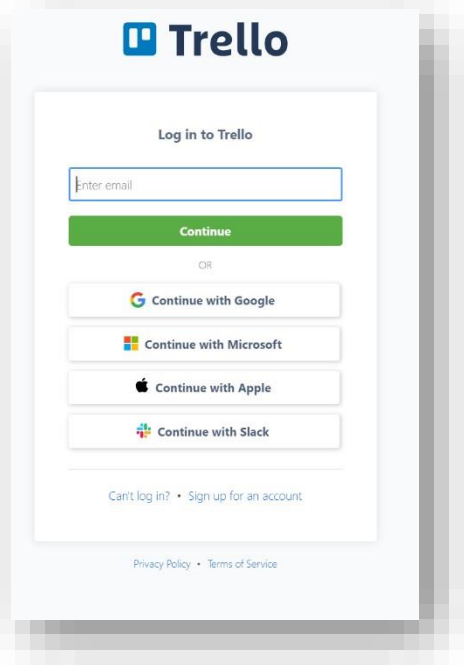
WeatherWell

Interface showcases a clean, minimalistic design with a natural colour-palette that is designed to be easy on the eyes. Whilst minimalistic WeatherWell manages to convey an immense amount of information from a single glance, from this screenshot we can see the upcoming weather as well as the highest peak its predicted to have, the overall risk level for the person and the chance of them having any ailments are on the bottom of the screen as well as the option to log how the user is feeling at that moment to fine-tune the apps ability to predict.



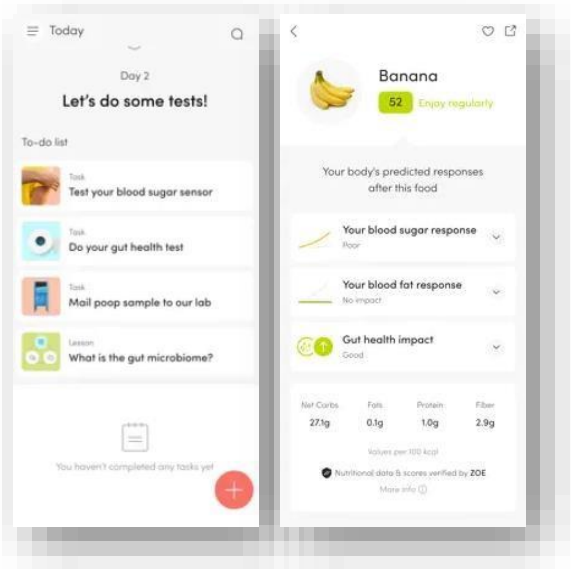
Trello

Trello's options to sign in stand out because of the unique feature that the user has where they can sign in using another platforms account. This functionality allows users who may be sceptical of Trello to sign-in with 100% confidence that there passwords will be secure, due to them being owned by thirdparty companies that will prevent the data from being accessed in the case of a breach.



Zoe

Zoe is a health app that is designed to help its users manage and control their overall health and wellbeing through health tracking, in order to do this the app has a wide range of features, the first of these is a personalised health plan. This allows users to set goals and track their progress towards achieving them. The app will also expand upon this by providing personalised advice based on their individual needs, for example if a user is diabetic then their fitness plan can be made with that in mind. The second key feature of Zoe is its ability to connect its users with professionals. This make the service more trustworthy for the users as well as being an incredibly useful alternative for people who may not have access to regular medical care.



However whilst Zoe can provide these services its original/ main purpose is to be used as a health study, gathering data that can be used to notice patterns or trends with certain issues users might be facing, whilst this is not a downside it does mean that their primary focus is not on the users themselves, I believe this is something I should be wary of when developing Health Advice Group's application and make sure the user is aware that they are the services number one priority.

MyAsthma

MyAsthma is a tool that is designed to help people with Asthma control their seasonal illness. My Asthma's main way of doing this is by utilising the latest newly emerging technologies and recent development in artificial intelligence and machine learning. They do this by using artificial intelligence to monitor any symptoms a patient may have, what kind of medication they take, as well as the number of times they puff their asthma inhaler a day. Using this information, a custom plan designed specifically for them can be made, taking in each patient's unique requirements

Overall the main key thing that sets MyAsthma apart from other health services is its use of newly emerging technologies, something that I will be able to keep in mind when developing the Health Advice Group service.

Source table:

Greenpeace	https://www.greenpeace.org.uk/
MyFitnessPal	https://www.myfitnesspal.com/
WeatherWell	https://www.weatherwell.app/
Trello	https://trello.com/
Zoe	https://joinzoe.com/
MyAsthma	https://mymhealth.com/

Activity A (ii)

A description of the proposed solution/ justification:

Health issues are more prevalent within society than ever before, with many people suffering from issues caused by the environment around them or allergies they were born with, because of this I've been contacted by the trustees of a charity called Health Advice Group whom want me to help tackle these issues by developing a digital solution that will provide:

- Advice on how to deal with extreme weather conditions
- Information on environmental health conditions and seasonal illnesses
- Risk assessments for home environments
- Forecast upcoming weather
- Access to a dashboard for monitoring air quality data
- Advice on how to deal with health matters affected by weather and environmental conditions

The digital solution I have decided on in order to facilitate the requests of the trustees of Health Advice Group is a website, this will provide an easy-to-use place where the user can track the environmental conditions and how they might affect their seasonal allergies.

When the user access the website for the first time they will be greeted with a login page, this page acts as a gateway between the user and the core-website as only members are allowed to continue,

signing-up to the site will be free and if the user doesn't have an account already then they will be able to create one using the registration page that will be linked in the pages navbar. This idea of a login page came from many applications that need to be able to store the user's data in a secure way, but I especially took inspiration from a website called Trello due to the added functionality of being able to login through an API that connects to a site like Google or Facebook, I have decided to incorporate this into the solution as it provides a fast and easy-to-use login experience. This fulfils the needs of the user as they can sign in without worry about their data's security being breached.

After successfully logging into the application the user will be redirected to the home page of the solution. This page acts as a dashboard, connecting every page together through a series of easily identifiable and simple buttons that once pressed will take you to the corresponding page. However it also provides an overview of the entire solution so the users don't have to go into specific pages in order to find out one piece of information that they need, this not only saves time but applies a minimalistic feel to the website as information is not cluttered but instead carefully placed so there is enough blank space between words so the context is still clear to the user. Things the dashboard will provide an overview of are elements like the current air quality and the type of weather, as well as displaying the users overall health so they can be aware how vulnerable they are to any risks. This fulfils the request of the client for a dashboard, I thought of this idea after the application MyFitnessPal as the key information is immediately available to the user upon opening the app, as referenced in the appendix above.

The next page expands upon the information from a specific widget in the home page, as it is dedicated to the weather forecasting to help inform the user of any health decisions. This page may be accessed through a button aptly named "weather forecasting" in the dashboard or through the widget displaying the current weather. In this page the same features the widget in the home page has are also displayed here but with added functionality. The user can change their location to view other places locations and can see the current temperature as well as have access to a panel that displays the predicted weather forecasts in the future. Additionally, in order to tie this page together and provide health advice based on the weather conditions an indicator will be placed in an easily viewable place on the page that will show the risk of the current environment. If the risk is deemed as unsafe, the user will receive as notification as well as advice on what to do next. I gained inspiration from the application WeatherWell due to their minimalistic and easy to understand approach when displaying the user's susceptibility to harsh weather conditions and decided to try and follow a similar route.

Next, Health Advice Group have requested a page dedicated to monitoring the air quality, in order to meet this request the website will incorporate this by displaying the AQI (air quality index) for the users location, this like on the weather page will provide the ability for the user to change where there location is listed as well so they can see the air quality of other locations. Expanding on this I've decided that a useful feature would be to include a graph that logs the previous quality of the air and displays it on a chart. This allows the user to recognise patterns the air quality has and, for example, if there is an incoming decline in the graph the user can prepare for it by moving location until it is deemed safe. A way to use images to convey this information to the user will be through a map where the air quality is represented using colours, purple representing the unhealthiest and green representing safe areas. I plan to use an API to achieve this. This was inspired through a charity called GreenPeace as their website helped me understand the importance of providing relevant and up-to-date information, links to GreenPeace are in the sources table above.

Occupational Specialism Task 1 – Proposal

Using the data from the previous two pages a system to calculate the users overall risk of environmental issues may be calculated. This page will also take into account data specific to the user such as their allergies making the end result completely tailored to them specifically. They'll input this data through a series of clean and minimalistic buttons that the user will click if they apply to them, such as hay fever. Once the risk has been calculated, the website will check the user's location to determine the best possible advice for them to receive. This advice will then be output for the user to read. This page can also double as a personal health tracking tool due to the users being able to log their health overtime through a input field and notice any changes overtime. I took inspiration from the application Zoe as they provide the ability for users to log their health in a similar way as seen in the appendix above.

I have also decided to add an extra page in order to help meet the needs of the users using the solution, this will be achieved through a help section where any issues or problems they have can be entered into a form that will be emailed to the trustees. This additional feature supports the users' needs by allowing them to communicate with the owners and in order to ensure the security of the data transmission from the website to their email all the data will be encrypted using an asymmetric key.

The trustees have also requested "accessibility features" to be incorporated into the solution. Accessibility is a key part as it will allows many users regardless of technical knowledge or ability to use the service, this will lead to an increase in the overall users of the solution thus resulting in a greater success for Health Advice Group. The way I intend to make the digital solution more accessible is by incorporating the FURPS model to improve how our system will work.

FURPS stands for 5 things, functionality, usability, reliability, performance and scalability, below I have gone into further detail on each of these points stating how they will be used within the system.

- Functionality, the digital solutions code will be clear and efficient as well as include comments for the purpose of every function, this way in the future if any new features are added or if old features are changed it will be easy to navigate where the code should be changed/ improved.
- Usability, the digital solution will provide a settings menu that will allow its users to customise how certain elements are displayed, for example if a user has difficulty seeing they can increase the size of the text or if they have dyslexia they can only see certain colours clearly they will be able to change the colour palette of the website. This makes the website more accessible to users who may have otherwise difficulty using the service and make the solution more inclusive and therefore complying with the WCAG (web accessibility) legislation
- Reliability, the solution will be accessible at all times through having a server always available to maintain the back-end of the solution so that regardless of when the user is free they will always be able to access the solution without any trouble, also if any errors with the solution are discovered they'll be quickly resolved by a programmer, so we do not breach the trust of our clients to provide a quality service.
- Performance, the developed solution will run quickly and have low loading times, this will be achieved by writing efficient code that only does what it needs to do in order for the needs of the user to be met as well as having enough servers to handle the load of the clients using the service

- Supportability, the developed solution will be accessible on multiple devices such as mobile, desktop, mac, etc. this is so people can access the solution from many different places and allowing the user to choose the version most suited to them at the moment they are using it, for example mobile is they are on the go. The way this will be implemented into the

  solution in order to prevent warped scaling due to different devices resolutions is through the use of viewports, these will detect the resolution of the device the user is currently on and send the appropriate version of the solution to them without any overlapping elements, this will provide a seamless experience that will prevent inconsistencies with the quality of the service.

To conclude, the above points successfully meet all of the requirements asked for by the Health Advice Group providing an accessible, minimalistic, easy-to-use digital solution that's unique take on its features will allow for easy brand recognisability.

Functional and non-functional requirements:

Functional Requirements			
No.	**Features**	**Justification**	**Priority**
1	Make use of a geolocation API to discover the users location	This is a necessary building block in the application as it will be used in order to provide relevant information about the upcoming weather, temperature and air quality of the user's area.	HIGH
2	All users must create an account in order to use the service	Allows the website to store data that is only relevant to the person that own the account	HIGH
3	Upon creation of an account an email will be sent to the users email address	Ensures that the user is creating an account with an email that they own, adding a layer of security to the solution	Low
4	Providing a form for the user to input all of their health conditions and how susceptible they are to the environment around them	This will be used when the system is determining whether or not to notify the user about any extreme conditions coming up, for example if the user has asthma then they will be notified sooner about any changes to the air quality that might be detrimental to their health	Medium
5	Display the upcoming weather of the users location	Informs the user about any issues they might have that are out of their control so they can plan around them, for example if a user can see there is heavy snow on Monday they'll be able to cancel work due to unsafe driving conditions.	Medium
6	Alert the user if there are any upcoming extreme weather conditions	This features purpose is to detect extremely unsafe weather conditions and inform the user about the severity of them to ensure that they will be able to remain safe, for example if there	HIGH

		is a tornado heading towards the users location they will be able to avoid it before it gets there.	
7	Display the upcoming temperatures of the users location	Allows the user to remain informed and can be used to help determine how many layers of clothing they should wear, etc.	Low
8	Alert and provide the user with advice on dealing with any extreme upcoming weather temperatures	In the case of sudden drastic changes in temperature, possibly caused by global warning, the user should be notified so they can aptly prepare, for example in the event of a heatwave they can ensure they have systems in place to	HIGH
		help them cool down such as having air conditioning	
9	Provide access to a dashboard that showcases the current air quality for the users area as well as the surrounding areas	This will allow the users to see the quality of the air of their location and allows them to determine whether or not it is acceptable. If the user is not satisfied with the air quality then the system will be able to provide them with advice on what actions they should take.	HIGH
10	Alert the user if the air quality is below the defined standard for 'healthy'	The air quality index is used to measure the air quality of a certain area, the lower the number used to describe the AQI the healthier the area is, for context anything above an AQI of 200 is deemed as poor, the solution should determine the severity of the AQI and present advice on what the users next steps should be, for example purchasing an air purifier, moving to another location,	Medium

Non-functional Requirements			
No.	**Features**	**Justification**	**Priority**
1	Availability at all times	Users will be able to access the solution whenever they are available, making it more inclusive to those who are busy or live in different time zones increasing the solutions reliability	Medium
2	All features being organised within a dashboard	This increases the accessibility of the solution as users will be able to locate to the desired page with ease and will need minimal technical knowledge to achieve this.	Medium

3	All images on the system will have alternate text for users using text-to-speech	This increases the usability of the solution as anyone with visual impairments, blindness or any other condition affecting their eyesight will be able to understand the context of the webpage with clarity	HIGH
4	Having code be concise and written as efficiently as possible	This will increase the systems performance by decreasing the number of ticks taken for the applications loading times and only loading what is necessary for the application to work as intended	Low
5	Scale correctly to the devices resolution	Different users will access the digital solution on different platforms, these platforms have different sized screens meaning that if this step is not done correctly certain elements of the page may overlap, etc. this can negatively affect the systems readability/ usability as the page might not work as intended and key information might be hidden from the user	HIGH
6	Having code be written with consistently styled variables	This makes the solution more maintainable as if there's ever an error or if Health Advice	Medium
	and regular comments to explain every function	Group wish to add more features any reasonably competent programmer should be able to look at the code and be able to understand/ modify it as they wish.	
7	Enable users to change the colour contrast of the solution	This makes the solution more accessible to users with colour-blindness or those sensitive to light meaning they'll be able to have a higher quality experience with the service	Medium

Decomposition of the problems that will need to be solved to implement the functional and non-functional requirements:

In order to implement the features and pages mentioned within my proposition there are several problems that will need to decomposed first, below I have written how I plan to decompose the problems and how they make the website have all the function and non-functional requirements requested by Health Advice Group.

One of the features requested by Health Advice Group that will be included in the solution is the ability to forecast the weather to inform health decisions of the users, for example if it's raining then they should stay inside to avoid getting wet. The first issue to be solved that occurs is finding out where the user currently is as the weather for that area cannot be displayed if the area is unknown, the way the solution will find this information is through the use of a geolocation application programming interface or API. This will act as a software intermediary between the solution and the application holding the location data and allow communication between the two systems. This then will allow the users current coordinates to be sent to the application so that it may be used for the specific function of displaying the upcoming weather. The next step to the desired result is through

the use of a second API that's goal is transmitting a reliable source of data regarding upcoming weather conditions back to the application.

Once the application has received the data, it can be put into context through the front-end of the web-designing process, through using HTML for the skeleton of the website, CSS to style the weather and JavaScript for any client-side logic such as making the text blue when there is upcoming snow to provide a visual cue for the user.

The next feature for the digital solution will make use of the geolocation API we have in place to display the air quality for the users location. The data regarding the air quality will also be able to be found using an API and once transmitted to the application will be able to be displayed on the website using the same front-end process for the weather forecasting with slight modifications so it's more specific to the air, such as showing the surrounding areas air quality instead of upcoming air quality predictions for the same area so it is more relevant to the user.

In order to be able to store the user's location and other information the system will need to be able to remember each unique user's information, we can develop this though a registration system if they have never been to the page before, or if they're a returning member they will be granted with a login screen. The way the system will be able to tell which one they are will be through a cookie pop-up when users access the website, if they click accept information about their visit will be able to be stored, if they decline their use of the system won't be affected in any way but next time they may have to manually navigate to the login page. Once a user has registered a function in the program will check their inputs, for example to see if their password is at least 8 characters, and decide whether or not it is acceptable. If all of the condition are met the users credentials can be saved to a database at the back-end of the solution, similarly if a returning user is logging in instead of registering their data will be run through the user database to see if their login matches any fields in the same row, if they don't they may try again, but if it does they can access the rest of the solution. The way our application will be able to transmit this data from the front-end and communicate with the back-end will be through the use of structured query language or SQL, this is a programming language that is designed for the management of databases and will allow fast and reliable transfer of data from the database.

Furthermore, another feature digital solution will provide is giving advice on how to deal with health matters affected by weather and environmental conditions, this will be achieved through the use of a form that will collect data about the user and any health issues they have, for example if they are asthmatic. Following this the system will be able to use the data it has collected on the air quality and weather to see if this is aiding their problems or will impact them negatively, if an issue is detected the solution will display a warning message to the user detailing what might be worsening their condition, for example if the user is asthmatic and they live in area with a very bad air quality then this will likely have an effect on their ability to breath. Expanding on this in order to give advice on what the user should do next the system will sort through a series of pre-existing responses to their ailment. For example, if the user is in the situation above the system would recommend purchasing an air purifier, making sure they take their inhaler when they should or move out of the area to one with a better air quality.

Finally, in order for the digital solution to be able to mitigate any risks or breaches in its security, numerous protocols and responses will be developed to combat these threats. An example of this is through encrypting all data stored within the databases such as the client's username or password, preventing SQL injections or man-in-the-middle attacks from having any effect on the users. Another system that will be put in place to prevent threats is the use of 2-factor-authentication to prevent

phishing attacks that don't allow foreign devices to log into their account without inputting a code received by the actual owner to a system that only they have3 access to, such as their email or their mobile phone.

KPIs (Key Performance Indicators):

Key-Performance indicators	Justification
How many users make an account compared to the number of people that visit the website	Keeping track of this will allow you to determine how many users are being lost from the websites initial impressions and if it seems like a large number of users then changes can be made to better gage their interest
The average amount of time users spend using the service	Keeping track of this will provide insight into how useful people find the service
Which pages of the website receive the highest number of visits	This will show the most popular features of the service and thus allow them to be prioritised for future updates as they are the most relevant to the users
The percentage of users that change the sites settings to make it more accessible and what they change	This will allow patterns to be recognised between users and thus the website may be tweaked accordingly to provide a higher quality experience, for example if the majority of users increase the text size then the increase will be set as the new default
How many new visitors join the website	Allows you to determine if the websites traction is increasing and makes it possible to predict the websites rate of growth

UAC (User Acceptance Criteria):

The user acceptance criteria decides how easily a new solution can be integrated amongst users daily lives, if the application doesn't meet the decided requirements then Health Advice Group will miss key groups of people that are integral to the success of the solution.

These requirements can be seen listed below:

- The design must be simple and minimalistic, this is so anyone without any deep technical knowledge will still be able to access the website
- If the user forgets their password they will be able to reset it via an email sent to their account
- Buttons must be bright and clear in order to ensure that the user can easily identify what they are used for
- Users must be able to navigate to the settings tab and be able to customise their version of the application with relative ease in order to ensure it is accessible to many different kinds of users
- The website will be able to adapt to multiple screens resolutions, so the user can use the solution regardless of what device they are using

How potential risks will be mitigated

Threat	Solution	Justification	Priority level
Phishing attacks	Ensure all new sign-ups to the service have to undergo two-factor-authentication	This prevents attacks from logging into accounts that they don't own even if they have the users credentials as it sends an additional code to an application/ device that only the user themselves have access to, e.g. their email	HIGH
Man-in-themiddle attacks	All data stored within the databases will be asymmetrically encrypted	This turns all sensitive data into gibberish that the intercepting attacker will not be able to understand, using asymmetric encryption will also ensure a much high level of security as it involves the use of a private key that the attacker has no way of knowing	HIGH
SQL Injection	Input validation, meaning all inputs received from the user undergo checks to ensure they are suitable to be sent through the system	Prevents users who are not authorised to view sensitive data from manipulating back-end databases to access information that was not intended to be displayed	HIGH
Distributeddenial-ofservice attacks (DDOS)	Having a warm site run simultaneously with the main site	A warm site is an exact replica of the main site, that has all of the same features and capabilities but none of the data from the databases, this increases the reliability of the service as in the case of a DDOS attack or something similar the application will be able to swap out with the warm site whilst it is repaired, the advantages of a warm site compared to a cold or hot site is it requires little effort to maintain as the core code is the exact same as the original, the only time spent will be transferring the data to the new temporary system.	HIGH

Data corruption	Have a backup of all user data using RAID 5	In the worse-case scenario of a system failure involving the deletion of all data from the databases the system will simply restore the data from using it's other hard drives containing the data or if the data has been deleted beyond simply sending data from the back-ups the parities in the RAID 5 model can be used to reconstruct the lost data, increasing the solutions reliability	HIGH
Scaling the application to varying resolutions incorrectly	Detect the viewport of the users screen and have the solution change accordingly	If the solution does not adapt correctly to the screens resolution certain elements may be obstructed or information hidden from the user, this will have an effect on the readability and functionality of the system and therefore must be resolved to ensure a consistent quality for our users	HIGH
Multiple users having the same login username	When signing up to the solution the users credentials should be compared to the login database and make sure their details are unique to them and provide an error message if not	This is important as it will prevent errors in the future, or prevent someone gaining unrestricted access to an account that doesn't belong to them to manipulate as they please and thus increasing the security of the solution	HIGH
Incorrect or out of date information about the user	Allowing users to access their profile and make changes to their personal information to ensure that the solution remains up-todate	Having incorrect/ out of date data could potentially impact the solutions ability to provide personalised advice on the users health matters, providing the ability to change this information will ensure that the application will be able to share relevant information with its user	Medium
Forgotten login information	In the event that a user has forgotten their login they will be able to send a reset link to their email to change the field that they have forgotten to something new	If a user cannot log into their account then they will be unable to use the solution causing the risk of losing the number of users of the service, providing this functionality will help to prevent that	HIGH

Insecure APIs	Thorough research into the APIs that are being used in the solution to ensure they have good security and are not prone to being targeted for attacks	If an API is insecure it means that people with malicious intent may be able to breach its security and access data that they weren't permitted to view, through ensuring the developers practice good API 'hygiene' this can be prevented	Low

How relevant regulatory guidelines and legal requirements, in relation to software development and the health sector will be addressed.

Guidelines	Purpose	How it will be addressed
General Data Protection Act GDPR 2016	Ensure personal data is protected and not used for malicious means	All personal data collected by the website will only be relevant to the functionality of the website, no personal data will be kept for longer than it needs to be, for example if a user deletes their account and the user will be able to have access to their data to ensure that it is up-to-date
Equality act of 2010	Prevent discrimination and ensure inclusivity	Accepting a terms of service once signing up to the website that states the user is not allowed to discriminate against anyone whilst using the site
WCAG 2.0	Ensures a shared standard for accessibility that meets the needs of anyone regardless of their ability	Users will be able to change how the website is displayed for them through a settings menu that will provide many features designed to make the site more accessible. There will also be alt-text provided for any image so users who use text-to-speech will still be able to use the service.
Display Screen Equipment (DSE)	Protecting users from health risks when working with equipment	Accepting a terms of service once signing up to the website that informs them of proper technique when using a computer, such as making sure the top of the monitor is eyelevel